

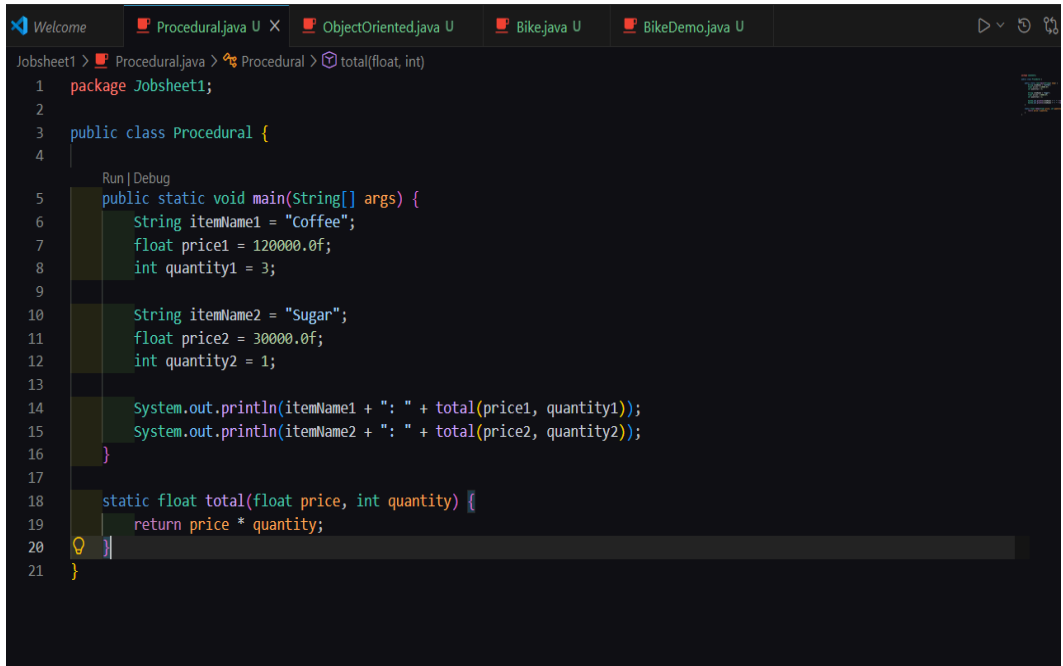


Mata Kuliah : Praktikum Pemrograman Berbasis Objek
Program Studi : D4 – Teknik Informatika
Semester : 3

Kelas : TI-2D
NIM : 254107020019
Nama : M. Javier Thufail
Jobsheet Ke- : 1

Laporan Jobsheet

Pendahuluan

Langkah	Jawaban/Deskripsi
Procedural	 <pre>1 package Jobsheet1; 2 3 public class Procedural { 4 5 Run Debug 6 public static void main(String[] args) { 7 String itemName1 = "Coffee"; 8 float price1 = 120000.0f; 9 int quantity1 = 3; 10 11 String itemName2 = "Sugar"; 12 float price2 = 30000.0f; 13 int quantity2 = 1; 14 15 System.out.println(itemName1 + ": " + total(price1, quantity1)); 16 System.out.println(itemName2 + ": " + total(price2, quantity2)); 17 } 18 19 static float total(float price, int quantity) { 20 return price * quantity; 21 } 22 }</pre>

Object Oriented

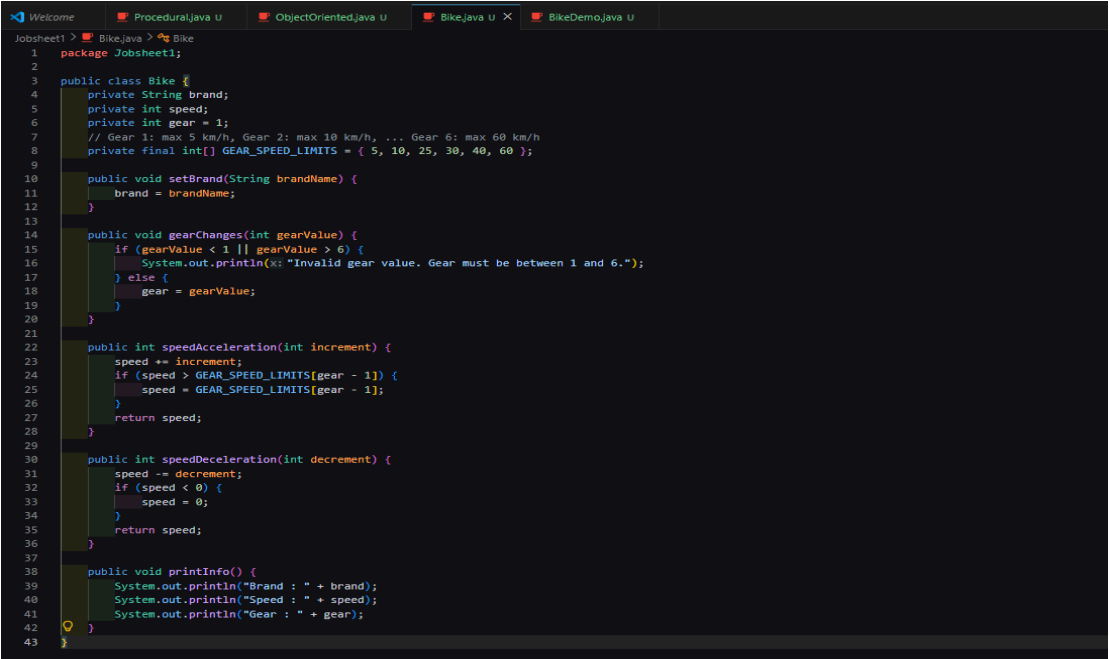
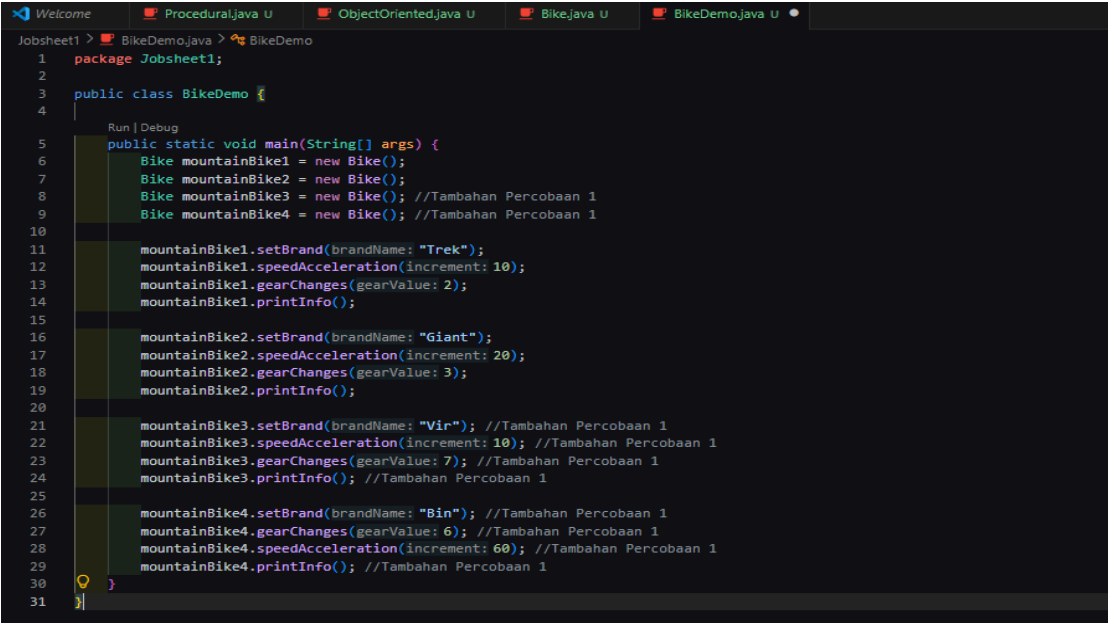
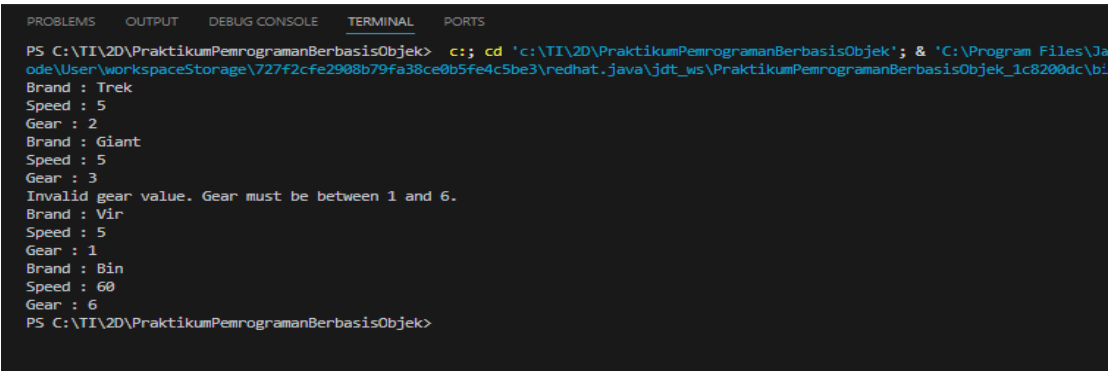
```
Jobsheet1 > ObjectOriented.java > ObjectOriented
1 package Jobsheet1;
2
3 class Product {
4     private final String itemName;
5     private final float price;
6     static int totbrg=0; //Tambahan
7
8     public Product(String itemName, float price) {
9         this.itemName = itemName;
10        this.price = price;
11        totbrg++; //Tambahan
12    }
13
14    public float total(int quantity) {
15        return price * quantity;
16    }
17
18    public String getName() {
19        return itemName;
20    }
21
22    public static void totbelanjaan() {
23        System.out.println("Total belanjaan = "+totbrg); //Tambahan
24    }
25 }
26
27 public class ObjectOriented {
28     Run | Debug
29     public static void main(String[] args) {
30         Product product1 = new Product(itemName: "Coffee", price: 120000.0f);
31         Product product2 = new Product(itemName: "Sugar", price: 30000.0f);
32         Product product3 = new Product(itemName: "Tea", price: 50000.0f); //Tambahan
33
34         System.out.println(product1.getName() + ": " + product1.total(quantity: 3));
35         System.out.println(product2.getName() + ": " + product2.total(quantity: 1));
36         System.out.println(product3.getName() + ": " + product3.total(quantity: 1)); //Tambahan
37         Product.totbelanjaan(); //Tambahan
38     }
39 }
```

Output

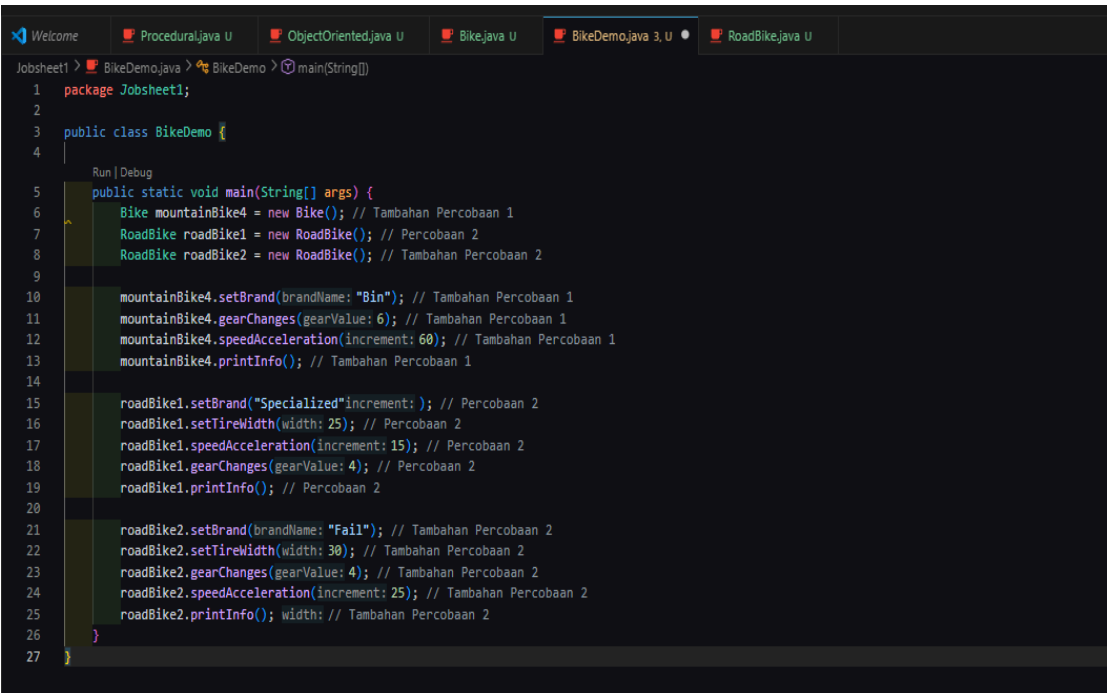
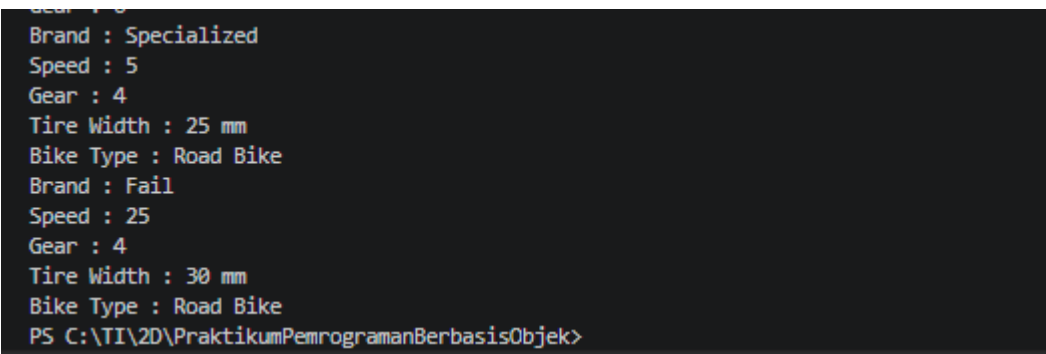
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\TII\2D\PraktikumPemrogramanBerbasisObjek> & 'C:\Program Files\Java\jdk-24\bin\java.exe'
c5be3\redhat_java\jdt_ws\PraktikumPemrogramanBerbasisObjek_1c8200dc\bin' 'Jobsheet1.ObjectOri
Coffee: 360000.0
Sugar: 30000.0
Tea: 50000.0
Total belanjaan = 3
PS C:\TII\2D\PraktikumPemrogramanBerbasisObjek>
```

Percobaan 1

Langkah	Jawaban/Deskripsi
Bike	 <pre> 1 package Jobsheet1; 2 3 public class Bike { 4 private String brand; 5 private int speed; 6 private int gear = 1; 7 // Gear 1: max 5 km/h, Gear 2: max 10 km/h, ..., Gear 6: max 60 km/h 8 private final int[] GEAR_SPEED_LIMITS = { 5, 10, 25, 30, 40, 60 }; 9 10 public void setBrand(String brandName) { 11 brand = brandName; 12 } 13 14 public void gearChanges(int gearValue) { 15 if (gearValue < 1 gearValue > 6) { 16 System.out.println("Invalid gear value. Gear must be between 1 and 6."); 17 } else { 18 gear = gearValue; 19 } 20 } 21 22 public int speedAcceleration(int increment) { 23 speed += increment; 24 if (speed > GEAR_SPEED_LIMITS[gear - 1]) { 25 speed = GEAR_SPEED_LIMITS[gear - 1]; 26 } 27 return speed; 28 } 29 30 public int speedDeceleration(int decrement) { 31 speed -= decrement; 32 if (speed < 0) { 33 speed = 0; 34 } 35 return speed; 36 } 37 38 public void printInfo() { 39 System.out.println("Brand : " + brand); 40 System.out.println("Speed : " + speed); 41 System.out.println("Gear : " + gear); 42 } 43 } </pre>
Bike Demo	 <pre> 1 package Jobsheet1; 2 3 public class BikeDemo { 4 5 Run Debug 6 public static void main(String[] args) { 7 Bike mountainBike1 = new Bike(); 8 Bike mountainBike2 = new Bike(); 9 Bike mountainBike3 = new Bike(); //Tambahan Percobaan 1 10 Bike mountainBike4 = new Bike(); //Tambahan Percobaan 1 11 12 mountainBike1.setBrand(brandName: "Trek"); 13 mountainBike1.speedAcceleration(increment: 10); 14 mountainBike1.gearChanges(gearValue: 2); 15 mountainBike1.printInfo(); 16 17 mountainBike2.setBrand(brandName: "Giant"); 18 mountainBike2.speedAcceleration(increment: 20); 19 mountainBike2.gearChanges(gearValue: 3); 20 mountainBike2.printInfo(); 21 22 mountainBike3.setBrand(brandName: "Vir"); //Tambahan Percobaan 1 23 mountainBike3.speedAcceleration(increment: 10); //Tambahan Percobaan 1 24 mountainBike3.gearChanges(gearValue: 7); //Tambahan Percobaan 1 25 mountainBike3.printInfo(); //Tambahan Percobaan 1 26 27 mountainBike4.setBrand(brandName: "Bin"); //Tambahan Percobaan 1 28 mountainBike4.gearChanges(gearValue: 6); //Tambahan Percobaan 1 29 mountainBike4.speedAcceleration(increment: 60); //Tambahan Percobaan 1 30 mountainBike4.printInfo(); //Tambahan Percobaan 1 31 } 32 } </pre>
Output	 <pre> PS C:\YI\2D\PraktikumPemrogramanBerbasisObjek> c:: cd 'c:\YI\2D\PraktikumPemrogramanBerbasisObjek'; & 'C:\Program Files\Java\jdk-8.0.60\bin\java.exe' -classpath .\bin\BikeDemo.class Brand : Trek Speed : 5 Gear : 2 Brand : Giant Speed : 5 Gear : 3 Invalid gear value. Gear must be between 1 and 6. Brand : Vir Speed : 5 Gear : 1 Brand : Bin Speed : 60 Gear : 6 PS C:\YI\2D\PraktikumPemrogramanBerbasisObjek> </pre>

Percobaan 2

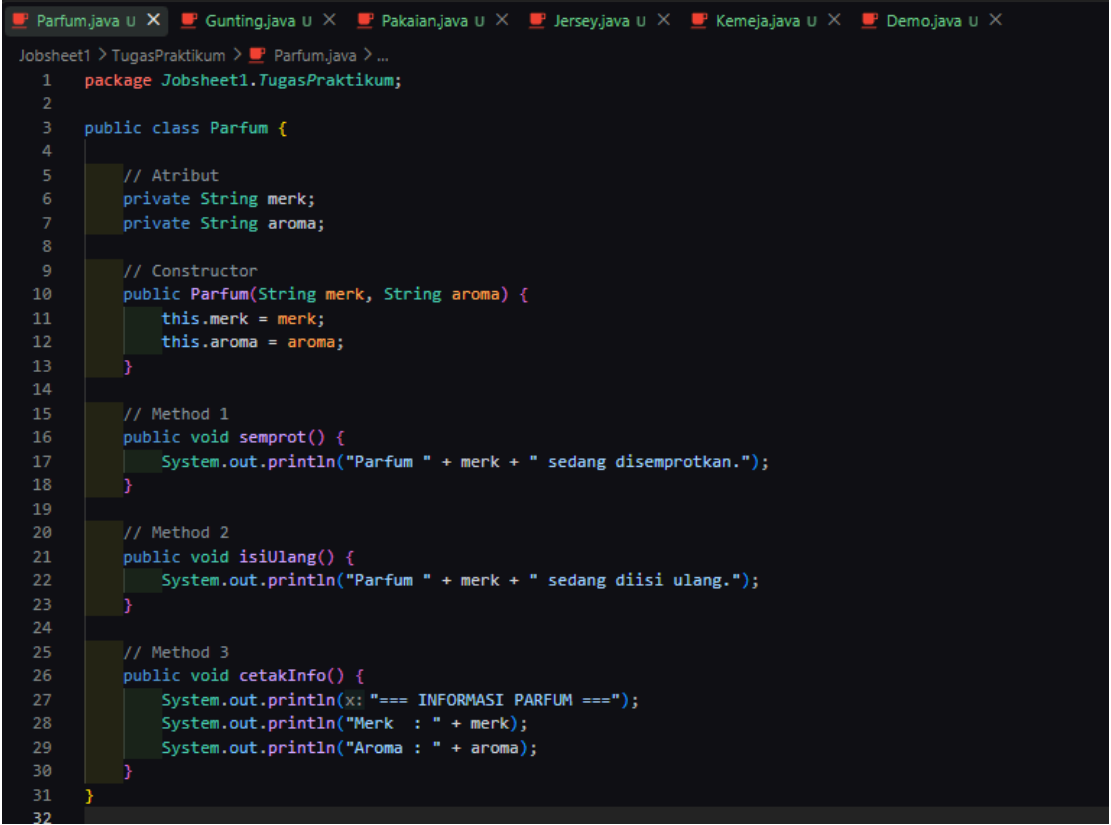
Langkah	Jawaban/Deskripsi
Road Bike	 <pre> 1 package Jobsheet1; 2 3 public class RoadBike extends Bike { 4 private int tireWidth; 5 6 public void setTireWidth(int width) { 7 tireWidth = width; 8 } 9 10 @Override 11 public void printInfo() { 12 super.printInfo(); 13 System.out.println("Tire Width : " + tireWidth + " mm"); 14 System.out.println("Bike Type : Road Bike"); 15 } 16 } 17 18 </pre>
Bike Demo	 <pre> 1 package Jobsheet1; 2 3 public class BikeDemo { 4 5 public static void main(String[] args) { 6 Bike mountainBike4 = new Bike(); // Tambahan Percobaan 1 7 RoadBike roadBike1 = new RoadBike(); // Percobaan 2 8 RoadBike roadBike2 = new RoadBike(); // Tambahan Percobaan 2 9 10 mountainBike4.setBrand("Bin"); // Tambahan Percobaan 1 11 mountainBike4.gearChanges(6); // Tambahan Percobaan 1 12 mountainBike4.speedAcceleration(60); // Tambahan Percobaan 1 13 mountainBike4.printInfo(); // Tambahan Percobaan 1 14 15 roadBike1.setBrand("Specialized"); // Percobaan 2 16 roadBike1.setTireWidth(25); // Percobaan 2 17 roadBike1.speedAcceleration(15); // Percobaan 2 18 roadBike1.gearChanges(4); // Percobaan 2 19 roadBike1.printInfo(); // Percobaan 2 20 21 roadBike2.setBrand("Fail"); // Tambahan Percobaan 2 22 roadBike2.setTireWidth(30); // Tambahan Percobaan 2 23 roadBike2.gearChanges(4); // Tambahan Percobaan 2 24 roadBike2.speedAcceleration(25); // Tambahan Percobaan 2 25 roadBike2.printInfo(); // Tambahan Percobaan 2 26 } 27 } </pre>
Output	 <pre> Gear : 6 Brand : Specialized Speed : 5 Gear : 4 Tire Width : 25 mm Bike Type : Road Bike Brand : Fail Speed : 25 Gear : 4 Tire Width : 30 mm Bike Type : Road Bike PS C:\TI\2D\PraktikumPemrogramanBerbasisObjek> </pre>

Pertanyaan Praktikum

No	Pertanyaan dan Jawaban
1	Jelaskan perbedaan antara object dengan class! Jawaban: Class adalah cetak biru (<i>blueprint</i>), sedangkan Object adalah wujud nyata (<i>instance</i>).
2	Jelaskan alasan gear dan brand dapat menjadi atribut dari object Bike! Jawaban: Karena keduanya merupakan karakteristik fisik yang mendeskripsikan ciri khas dari object bike.
3	Sebutkan salah satu kelebihan utama dari pemrograman berorientasi objek dibandingkan dengan pemrograman prosedural! Jawaban: Kelebihan PBO adalah kemudahan perawatan kode, jika ada perubahan di satu bagian tidak mudah merusak seluruh program.
4	Apakah diperbolehkan melakukan pendefinisian dua buah atribut dalam satu baris kode seperti “public String nama, alamat;”? Jawaban: Boleh. Karena tipe datanya sama
5	Pada class RoadBike, jelaskan alasan atribut brand, speed, dan gear tidak lagi ditulis di dalam class tersebut! Jawaban: Karena menerapkan Inheritance; RoadBike mewarisi kelas Bike.

Tugas Praktikum

No	Jawaban/Deskripsi
1	1. parfum. 2. Gunting.   3. Jersey 4. Kemeja  
2	1. Class Parfum Atribut : merk, aroma Method : semprot(), isiUlang(), cetakInfo() 2. Class Gunting Atribut : bahan, warnaGagang Method : potong(), asah(), cetakInfo()

	3. Class Pakaian Atribut : bahan, ukuran Method : dipakai(), dicuci(), cetakInfo() 4. Class Jersey Atribut : namaPemain, nomorPunggung Method : cetakNomor(), cetakNama(), cetakInfo() 5. Class Kemeja Atribut : tipeLengan, jumlahKancing Method : kancingan(), disetrika(), cetakInfo()
3	Parfum.java  <pre> 1 package Jobsheet1.TugasPraktikum; 2 3 public class Parfum { 4 5 // Atribut 6 private String merk; 7 private String aroma; 8 9 // Constructor 10 public Parfum(String merk, String aroma) { 11 this.merk = merk; 12 this.aroma = aroma; 13 } 14 15 // Method 1 16 public void semprot() { 17 System.out.println("Parfum " + merk + " sedang disemprotkan."); 18 } 19 20 // Method 2 21 public void isiUlang() { 22 System.out.println("Parfum " + merk + " sedang diisi ulang."); 23 } 24 25 // Method 3 26 public void cetakInfo() { 27 System.out.println(x: "=== INFORMASI PARFUM ==="); 28 System.out.println("Merk : " + merk); 29 System.out.println("Aroma : " + aroma); 30 } 31 } 32 </pre>

4

Gunting.java:

```

Parfum.java U X  Gunting.java U X  Pakaian.java U X  Jersey.java U X  Kemeja.java U X  Demo.java U X
Jobsheet1 > TugasPraktikum > Gunting.java > Gunting
1  package Jobsheet1.TugasPraktikum;
2
3  public class Gunting {
4
5      // Atribut
6      private String bahan;
7      private String warnaGagang;
8
9      // Constructor
10     public Gunting(String bahan, String warnaGagang) {
11         this.bahan = bahan;
12         this.warnaGagang = warnaGagang;
13     }
14
15     // Method 1
16     public void potong() {
17         System.out.println(x: "Gunting digunakan untuk memotong.");
18     }
19
20     // Method 2
21     public void asah() {
22         System.out.println(x: "Gunting sedang diasah agar tetap tajam.");
23     }
24
25     // Method 3
26     public void cetakInfo() {
27         System.out.println(x: "=== INFORMASI GUNTING ===");
28         System.out.println("Bahan      : " + bahan);
29         System.out.println("Warna Gagang: " + warnaGagang);
30     }
31 }

```

5

Pakaian.java:

```

Parfum.java U X  Gunting.java U X  Pakaian.java U X  Jersey.java U X  Kemeja.java U X  Demo.java U X
Jobsheet1 > TugasPraktikum > Pakaian.java > Pakaian
1  package Jobsheet1.TugasPraktikum;
2
3  public class Pakaian {
4
5      // Atribut
6      protected String bahan;
7      protected String ukuran;
8
9      // Constructor
10     public Pakaian(String bahan, String ukuran) {
11         this.bahan = bahan;
12         this.ukuran = ukuran;
13     }
14
15     // Method 1
16     public void dipakai() {
17         System.out.println(x: "Pakaian sedang dipakai.");
18     }
19
20     // Method 2
21     public void dicuci() {
22         System.out.println(x: "Pakaian sedang dicuci.");
23     }
24
25     // Method 3
26     public void cetakInfo() {
27         System.out.println(x: "=== INFORMASI PAKAIAN ===");
28         System.out.println("Bahan : " + bahan);
29         System.out.println("Ukuran : " + ukuran);
30     }
31 }

```

6

Jersey.java:

```

Parfum.java u x  Gunting.java u x  Pakaian.java u x  Jersey.java u x  Kemeja.java u x  Demo.java u x
Jobsheet1 > TugasPraktikum > Jersey.java > Jersey(String, String, String, int)
1  package Jobsheet1.TugasPraktikum;
2
3  public class Jersey extends Pakaian {
4
5      // Atribut
6      private String namaPemain;
7      private int nomorPunggung;
8
9      // Constructor
10     public Jersey(String bahan, String ukuran,
11                   String namaPemain, int nomorPunggung) {
12
13         // Memanggil constructor class Pakaian
14         super(bahan, ukuran);
15
16         this.namaPemain = namaPemain;
17         this.nomorPunggung = nomorPunggung;
18     }
19
20     // Method 1
21     public void cetakNomor() {
22         System.out.println("Nomor punggung jersey: " + nomorPunggung);
23     }
24
25     // Method 2
26     public void cetakNama() {
27         System.out.println("Nama pemain pada jersey: " + namaPemain);
28     }
29
30     // Method 3
31     @Override
32     public void cetakInfo() {
33         System.out.println(x: "=== INFORMASI JERSEY ===");
34         System.out.println("Bahan      : " + bahan);
35         System.out.println("Ukuran   : " + ukuran);
36         System.out.println("Nama Pemain : " + namaPemain);
37         System.out.println("Nomor Punggung : " + nomorPunggung);
38     }
39 }

```

7

Kemeja.java:

```

Parfum.java u x  Gunting.java u x  Pakaian.java u x  Jersey.java u x  Kemeja.java u x  Demo.java u x
Jobsheet1 > TugasPraktikum > Kemeja.java > Kemeja
1  package Jobsheet1.TugasPraktikum;
2
3  public class Kemeja extends Pakaian {
4
5      // Atribut
6      private String tipeLengan;
7      private int jumlahKancing;
8
9      // Constructor
10     public Kemeja(String bahan, String ukuran,
11                   String tipeLengan, int jumlahKancing) {
12
13         // Memanggil constructor Pakaian
14         super(bahan, ukuran);
15
16         this.tipeLengan = tipeLengan;
17         this.jumlahKancing = jumlahKancing;
18     }
19
20     // Method 1
21     public void kancingkan() {
22         System.out.println(x: "Kemeja sedang dikancingkan.");
23     }
24
25     // Method 2
26     public void disetrika() {
27         System.out.println(x: "Kemeja sedang disetrika.");
28     }
29
30     // Method 3
31     @Override
32     public void cetakInfo() {
33         System.out.println(x: "=== INFORMASI KEMEJA ===");
34         System.out.println("Bahan      : " + bahan);
35         System.out.println("Ukuran   : " + ukuran);
36         System.out.println("Tipe Lengan : " + tipeLengan);
37         System.out.println("Jumlah Kancing : " + jumlahKancing);
38     }
39 }

```

8

Demo.java:

```

1 package Jobsheet1.TugasPraktikum;
2
3 public class Demo {
4
5     public static void main(String[] args) {
6
7         Parfum parfum = new Parfum();
8         parfum.setNama("bjorka");
9         parfum.setAroma("Desire Moment");
10    }
11
12    Gunting gunting = new Gunting();
13    gunting.setBahan("Stainless Steel");
14    gunting.setWarnaGagang("Hitam");
15
16    Pakaian pakaian = new Pakaian();
17    pakaian.setBahan("Katun");
18    pakaian.setUkuran("L");
19
20    Jersey jersey = new Jersey();
21    jersey.setBahan("Masuma");
22    jersey.setUkuran("L");
23    jersey.setNamaPemain("THUFAIL");
24    jersey.setNomorPunggung(15);
25
26    Kemeja kemeja = new Kemeja();
27    kemeja.setBahan("Katun");
28    kemeja.setUkuran("L");
29    kemeja.setTipeLengan("Pendek");
30    kemeja.setJumlahKancing(6);
31
32    System.out.println("=====");
33    System.out.println("OBJEK PARFUM");
34    System.out.println("=====");
35    parfum.sempotkan();
36    parfum.disiikan();
37    System.out.println();
38    System.out.println("=====");
39    System.out.println("OBJEK GUNTING");
40    System.out.println("=====");
41    gunting.digunakan();
42    gunting.diasah();
43    System.out.println();
44    System.out.println("=====");
45    System.out.println("OBJEK PAKAIAN");
46    System.out.println("=====");
47    pakaian.dipakai();
48    pakaian.dicuci();
49    System.out.println();
50    System.out.println("=====");
51    System.out.println("OBJEK JERSEY");
52    System.out.println("=====");
53    jersey.setNomorPunggung();
54    jersey.setNamaPemain();
55    System.out.println();
56    System.out.println("=====");
57    System.out.println("OBJEK KEMEJA");
58    System.out.println("=====");
59    kemeja.dikancingkan();
60    kemeja.dicuci();
61    System.out.println();
62 }

```

9

Outputnya:

```

PS C:\TI\20\PraktikumPemrogramanBerbasisObjek> & 'C:\Program Files\Java\jdk-8.0.0\bin' 'Jobsheet1.TugasPraktikum.Demo'
=====
OBJEK PARFUM
=====
Parfum bjorka sedang disemprotkan.
Parfum bjorka sedang diisi ulang.
=== INFORMASI PARFUM ===
Merk : bjorka
Aroma : Desire Moment
=====
OBJEK GUNTING
=====
Gunting digunakan untuk memotong.
Gunting sedang diasah agar tetap tajam.
=== INFORMASI GUNTING ===
Bahan : Stainless Steel
Warna Gagang: Hitam
=====
OBJEK PAKAIAN
=====
Pakaian sedang dipakai.
Pakaian sedang dicuci.
=== INFORMASI PAKAIAN ===
Bahan : Katun
Ukuran : L
=====
OBJEK JERSEY
=====
Nomor punggung jersey: 15
Nama pemain pada jersey: THUFAIL
=== INFORMASI JERSEY ===
Bahan : Masuma
Ukuran : L
Nama Pemain : THUFAIL
Nomor Punggung : 15
=====
OBJEK KEMEJA
=====
Kemeja sedang dikancingkan.
Kemeja sedang disetrika.
=== INFORMASI KEMEJA ===
Bahan : Katun
Ukuran : L
Tipe Lengan : Pendek
Jumlah Kancing: 6
PS C:\TI\20\PraktikumPemrogramanBerbasisObjek>

```